PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 1 • Chapter OL-T08 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

3 classified items • 7 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T08 • Expertise Q16+ • 3 items • 7 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Fractions, Decimals & Percentages

Unit 1 • Expertise • 7 marks • 3 item(s)

June 2025 | 4MA1-2H Q18 | 2 marks

Pure recurring block

November 2021 | 4MA1-1H Q18 | 3 marks

Non-recurring prefix followed by a recurring block

January 2023 | 2H Q16 | 2 marks

Decimal to fraction or percentage

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q18 **2 marks**

18 Use algebra to show that $0.\dot{7}0\dot{2} = \frac{26}{37}$

(Total for Question 18 is 2 marks)

WORKING SPACE

4MA1-2H Q18 **2 marks**

18 Use algebra to show that $0.\dot{7}0\dot{2} = \dfrac{26}{37}$

(Total for Question 18 is 2 marks)

Convert a recurring decimal to a fraction **2 marks**

Align a complete repeating block

$$x = 0.\overline{702}, \qquad 1000x = 702.\overline{702}.$$

Subtract and simplify

$$999x = 702 \Rightarrow x = \frac{702}{999} = \frac{26}{37}.$$

Final answer

$$\frac{26}{37}$$

November 2021 | 4MA1-1H | Question 18 | 3 marks

Chapter OL-T08: Fractions, Decimals & Percentages

18 $0.4\dot{x}$ is a recurring decimal.
x is a whole number such that $1 \leqslant x \leqslant 9$

Find, in terms of x, the recurring decimal $0.4\dot{x}$ as a fraction.
Give your fraction in its simplest form.
Show clear algebraic working.

....

Worked solution

November 2021 | Question 18 | 3 marks

Family: Convert a recurring decimal to a fraction • Variant: Non-recurring prefix followed by a recurring block

Method / working
1. Let
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
2. y=0.4\dot{x}
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
3. So
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
4. y=0.4xxx\ldots
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
5. Multiply by \(10\) and \(100\):
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
6. 10y=4.xxx\ldots
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
7. 100y=4x.xxx\ldots
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
8. Subtract:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
9. 100y-10y=(4x.xxx\ldots)-(4.xxx\ldots)
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
10. 90y=x+36
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
11. Therefore
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
12. y=frac{x+36}{90}
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
13. Answer: \(frac{x+36}{90}\)
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
Final answer: frac{x+36}{90}

January 2023 | 2H | Question 16 | 2 marks

Chapter OL-T08: Fractions, Decimals & Percentages

16 Use algebra to show that $0.4\dot{3}\dot{8} = \dfrac{217}{495}$

(Total for Question 16 is 2 marks)

Answer reference | January 2023 | Question 16

Family: Convert terminating representations • Variant: Decimal to fraction or percentage

Question	Working	Answer	Mark		Notes
16	eg $1000x = 438.38\ldots$ $\underline{\quad 10x = \quad 4.38\ldots}$ **or** $100x = 43.838\ldots$ $\underline{\quad x = \quad 0.438\ldots}$ oe		2	M1	For **selecting** 2 correct recurring decimals that when subtracted give a whole number or terminating decimal (43.4 or 434 etc) eg $1000x = 438.38...$ and $10x = 4.38....$ **or** $100x = 43.838\ldots$ and $x = 0.438\ldots$ with intention to subtract. (if recurring dots not shown then showing at least **one** of the numbers to at least 5sf) **or** $0.4 + 0.0\dot{3}\dot{8}$ **and** eg $1000x = 38.38...$ & $10x = 0.3838...$, with intention to subtract.
	eg $1000x - 10x = 438.38\ldots - 4.38\ldots = 434$ **and** $\frac{434}{990} = \frac{217}{495}$ **or** eg $100x - x = 43.838\ldots - 0.438\ldots = 43.4$ **and** $\frac{43.4}{99} = \frac{217}{495}$ **or** eg $1000x - 10x = 38.38\ldots - 0.3838 = 38$ and $0.4 + \frac{38}{990} = \frac{4 \times 99 + 38}{990} = \frac{434}{990} = \frac{217}{495}$ oe *working required*	Clearly shown		A1	For completion to $\frac{217}{495}$ dep on M1 and use of some algebra
					Total 2 marks

Worked solution

January 2023 | Question 16 | 2 marks

Family: Convert terminating representations • Variant: Decimal to fraction or percentage

Official final mark-scheme pages are embedded immediately after this question.